



Semantic Web Rule Language (from A Semantic Web Primer, Foundation of Semantic Web)

Knowledge Base and Reasoning

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- Reasoning berbasis Tableau Algorithm (logic, DL based, inferensi based on logic semantic)
Ex: FaCT, HermiT, Pellet
- Reasoning berbasis aturan (rule language), rule-based. Istilahnya R-Box
- Aturan eksplisit, horn-like rules atau logika predicate (First-Order Logic/FOL) sifatnya umum
Ex: Drools (di Protege)



- Proporsional $\rightarrow (x \rightarrow y)$
- Predicate $\rightarrow \text{hasChild}(x,y)$
- Individu $\rightarrow \text{Jim, John}$
- Variabel $\rightarrow x,y \dots$
- Kuantor $\rightarrow \text{Exist dan Universal}$
- Operator $\rightarrow \wedge (\text{dan}), \vee (\text{atau}), \neg (\text{tidak}), \rightarrow (\text{implikasi})$



- Disajikan dengan gunakan konsep OWL (classes, properties, individuals)
- Menggunakan notasi sequence (^) untuk semua rantai logical and operator
- Seperti rantai if ... then ..., yang kiri(body) harus True untuk terekseskusi rule menghasilkan fakta baru yg kanan (head)
- Menggunakan proses reasoning



- Dari ontology Keluarga, jalankan reasoner, akan terlihat fakta baru hasil (Inferred)

The screenshot displays the Protégé ontology editor with the following components:

- Class hierarchy: Man**: A tree view showing the hierarchy of classes. The 'Man' class is highlighted, and its subclasses are listed: Deceased, Human $\equiv$ Person, MyBDGuest, NarcisticPerson, Orphan, ChildLessPerson, Person $\equiv$ Human, ChildLessPerson, Parent, Father $\equiv$ Man, Grandfather, JohnsChildren, Man $\equiv$ Father, Grandfather, Mother $\equiv$ Woman, and Woman $\equiv$ Mother.
- Instances: Jack**: A list of instances for the 'Man' class. The instances are Jack, James, and Jim. Jack is selected.
- Annotations: Jack**: A panel showing the annotations for the selected instance 'Jack'.
- Description: Jack**: A panel showing the description of the selected instance 'Jack'. It lists the types: hasChild exactly 3 Person, hasWife exactly 1 Woman, Man, and Father.
- Property assertions: Jack**: A panel showing the property assertions for the selected instance 'Jack'. It lists the object property assertions: hasAge "53"^^xsd:nonNegativeInteger.



Active Ontology x Entities x Individuals by class x DL Query x Individual Hierarchy Tab x OntoGraf x ROWLTab x SPARQL Query x

Class hierarchy: Woman ? I = □ ×

Inferred

- Deceased
- Human ≡ Person
- MyBDGuest
- NarcisticPerson
- Orphan
 - ChildLessPerson
- Person ≡ Human
 - ChildLessPerson
 - Parent
 - Father ≡ Man
 - Grandfather
 - JohnsChildren
 - Man ≡ Father
 - Grandfather
 - Mother ≡ Woman
 - Woman ≡ Mother

Instances: Mary ? I = □ ×

For: Woman

Mary

Annotations Usage

Annotations: Mary ? I = □ ×

Annotations +

Description: Mary ? I = □ ×

Property assertions: Mary I = □ ×

Types +

- Woman ? @ × ○
- Mother ? @

Same Individual As +

Different Individuals +

Object property assertions +

- hasHusband John ? @
- hasSpouse John ? @

Data property assertions +

Negative object property assertions +

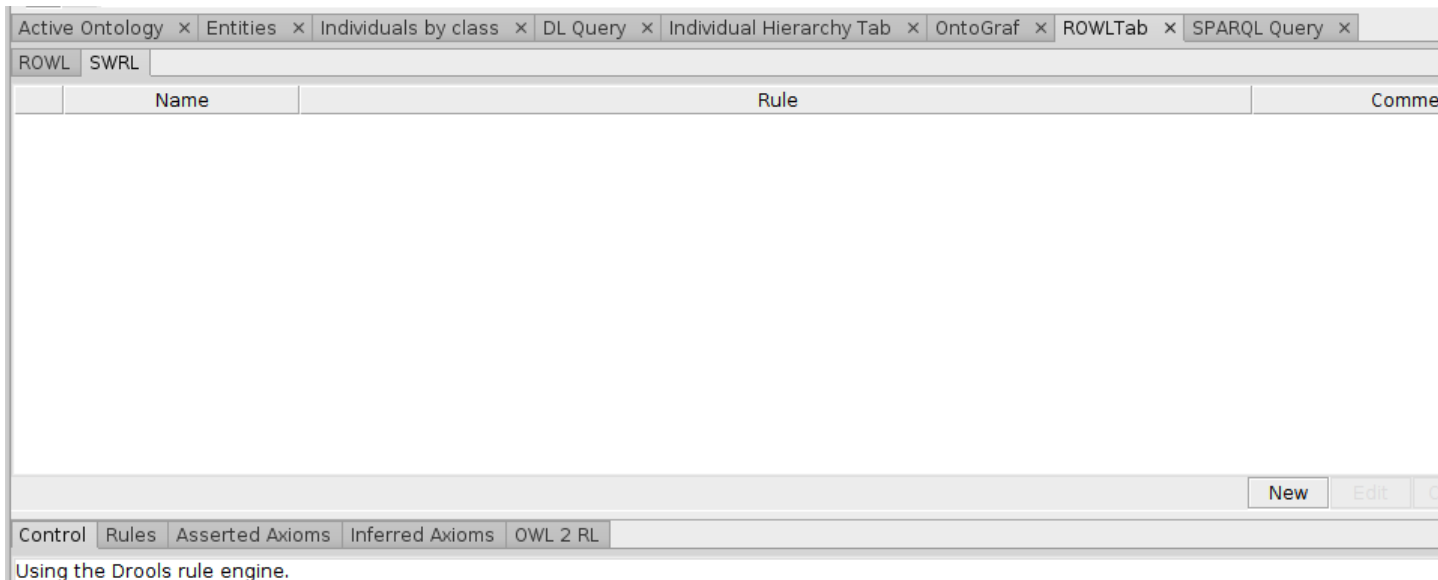
Negative data property assertions +



- Penambahan **rule dengan SWRL**, akan menghasilkan hal yang sama dengan hasil reasoning dari constraint ontology
- Ex: “hasChild” dengan “parentOf”
A “hasChild” B memiliki makna sama dengan A “parentOf” B
Knowledge ini **belum menjadi fakta eksplisit (implisit)** dalam Knowledge Base.
- Dengan reasoning ataupun SWRL dapat menjadi fakta baru



- Rule dapat ditambahkan lewat SWRL
- Ada **tab SWRL** → teknis bisa berbeda tergantung version of tools
- SWRL adalah format lain dari proses logic reasoning





- Ex: hasChild dan parentOf
- Dengan expressivity OWL 2 DL:
Diberikan constraint hasChild “Equivalent To” parentOf
- Dengan SWRL: diberikan aturan
 $\text{hasChild}(\text{?x}, \text{?y}) \rightarrow \text{parentOf}(\text{?x}, \text{?y})$

Note*:

- SWRL digunakan untuk menangani kompleksitas rule-based (tambahan) atau jika ingin cepat menggunakan (if – then)
- dapat bekerja bersama, Reasoning berbasis DL kemudian Drools



Di Protege

- OWL+SWRL->Drools : mengirim SWRL dan OWL ke the rule engine.
- **Drools** ini adalah rule engine tambahan dari protege, bekerja seperti forward chaining (expert system konvensional), namun tdk diukur konsistensinya
- Run Drools: menjalankan the rule engine.
- Drools→OWL: mengirimkan inferred knowledge to OWL



- Rule1:

$\text{hasParent}(\text{?p1}, \text{?p2}) \rightarrow \text{parentOf}(\text{?p1}, \text{?p2})$

Edit
Name
S1
Comment
Status
Ok
<code>hasParent(?p1,?p2) -> parentOf(?p1,?p2)</code>

- Rule2:

$\text{NarcisticPerson}(\text{?p}) \rightarrow \text{Person}(\text{?p})$



- Class $\rightarrow$ A-Box
Ex: Person(?p)
- Property $\rightarrow$ T-Box
Ex: hasParent(?p1,?p2), hasAge(?p1,?a)
- Rule $\rightarrow$ R-Box
 $\text{hasParent}(\text{?p1}, \text{?p2}) \rightarrow \text{parentOf}(\text{?p2}, \text{?p1})$
- **Tidak (belum) bisa negasi**, atau mengisi value, focus pada query-search dan classification/ labeling
- Bagus digunakan utk representasi knowledge tacit (ada di expert).



- Terdapat semacam **built in function** yang dapat dimanfaatkan dalam SWRL
- Primitive → komparasi, math, string, date dll
- Rule yg dituliskan adalah **bagian dari ontology**, sebagai tambahan atau meningkatkan reasoning
- Ex:
 - swrlb:greaterThan
 - swrlb:lessThanOrEqual
 - swrlb:add
 - etc



Further Reading

- Wardani, D., Jannah, R., Setiadi, H., 2020. Major Choosing Decision Support System with Talent Interest Approach Using SWRL and SAW Method, in: Proceedings of the 5th International Conference on Sustainable Information Engineering and Technology, SIET '20. Association for Computing Machinery, New York, NY, USA, pp. 95–100.
<https://doi.org/10.1145/3427423.3427445>
- <https://www.w3.org/submissions/SWRL/>